

Ethan Borkosky

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PROFESSIONAL SUMMARY

Full-stack software engineer with 6 years of professional experience building scalable web applications, prediction engines, and data-driven tools. Proficient across the entire development lifecycle from backend architecture and API design to complex, interactive frontends. Experienced with modern frameworks including Angular, React, Django REST, .NET, and Node.js, with a strong foundation in clean architecture, relational data modeling, and agile workflows.

TECHNICAL SKILLS

Languages	JavaScript / TypeScript, Python, C#, PHP, Golang, HTML/CSS
Frameworks	Angular, React, .NET, Django REST, Node.js, Flutter, Ionic, Laravel
Databases	MySQL, PostgreSQL, NoSQL, Firebase Firestore
Tools & Cloud	Git, GitHub, Docker, Digital Ocean, Firebase, Google Cloud Platform
Practices	Clean Architecture, REST API Design, Agile / Scrum, CI/CD

PROFESSIONAL EXPERIENCE

Software Engineer • **TalentValue Inc.** July 2020 – April 2026

Frontend: Angular 15, TypeScript, RxJS, NgRx, Ionic, d3.js, React (Vite / DaisyUI) | Backend: Django REST, FastAPI, Node.js, MySQL, Firebase

- Built and maintained a compensation intelligence platform (Angular 15 / Ionic / TypeScript) used to evaluate pay for thousands of roles across 17,000+ companies, leveraging real-time MarketValue prediction data.
- Led end-to-end development of the core MarketValue calculator ("Search"), the product's primary feature, architecting it as a tree of reactive Angular components with NgRx for predictable state management across candidate, company, valuation, location, market-space, and headcount inputs using @Input setters and RxJS streams to recompute predictions live as users adjusted criteria.
- Architected Market Tracker, a complex "smart spreadsheet" UI for team and department-level compensation benchmarking, implementing OnChanges-driven change detection, bulk-edit flows, and xlsx-based import/export, balancing high interactivity with clear data presentation.
- Implemented canActivate route guards enforcing both Firebase authentication and plan-based entitlement checks, gating feature access by user product tier across the app.
- Built a library of reusable custom d3.js data-visualization components (normal-distribution curves, gauges, comparative pay curves) and custom Angular pipes and directives to render prediction data and pay ranges consistently across the app.
- Designed and implemented server-side PDF report generation using Node.js and SendGrid, automating report delivery directly to users.
- Ingested and normalized public H-1B compensation and hiring range data from raw CSV datasets using Python, integrating them into the MySQL prediction model and measurably improving accuracy for individual contributor roles.
- Developed Node.js Firebase Cloud Functions supporting backend automation, including email workflows, report sharing, and server-side processing.
- Replaced a legacy prediction system by architecting a new Python/Django REST service with modular architecture, redesigned relational MySQL schemas, and optimized query performance for batch prediction workloads.

PERSONAL PROJECTS

Who's my Saint — Saint Matching App

Stack: *NestJS · Prisma · PostgreSQL (Supabase) · pgvector · Next.js · OpenAI Embeddings*

- Designed and built a public REST API in NestJS serving 151+ saints with tiered access control, separating rate-limited public endpoints from a protected admin boundary enforced via custom NestJS guard middleware restricting all write operations to authorized clients.
- Modeled a fully normalized relational schema in Prisma/PostgreSQL with join tables across patronages, tags, vocations, fun facts, and name variants, seeded programmatically from a structured JSON dataset.
- Engineered a weighted scoring algorithm in TypeScript that matches users to saints across five signals (name variant lookup, feast day proximity decay, vocation match, occupation-to-patronage keyword overlap, and feelings chip tag overlap) with per-signal normalization so disabled inputs don't skew results.
- Implemented pgvector on Supabase for semantic embedding-based candidate retrieval using OpenAI text-embedding-3-small, combining vector similarity pre-filtering with structured weighted scoring for optimal match quality.

Architected the API as a standalone deployable service with a versioned public surface, designed for future third-party developer access with usage tiers, rate limiting, and key-based registration.

Lumenform — Cohort management app for Catholic OCIA/RCIA faith formation programs

Stack: *.NET 9 Web API · Supabase (PostgreSQL + Auth) · Next.js · Mantine UI · React Native (planned)*

- Building a full-stack parish management tool as both a portfolio piece and a free service for Catholic communities.
- Implemented Clean Architecture with domain-driven models for cohorts, memberships, assignments, and events, along with JWT authentication and role-based access control.
- Developed a responsive Next.js dashboard with tabbed navigation and real-time API integration.

EDUCATION

Awesome Inc. Web Development Bootcamp *Lexington, KY*

16-week, full-time bootcamp covering the complete web development lifecycle — from pseudocode to deployment — with a focus on agile development practices.

Transylvania University *Lexington, KY*

Music Technology & Computer Science